

Nexora Corporate Knowledge Base: Architecture Standards

Target Role: ENGINEERING | Version: Official | Published: 2026-09-25

1. Architectural Paradigm

Nexora systems follow a decoupled, microservices architecture. Core backend services are written in Go for high-throughput concurrency, machine learning & analytics pipelines in Python 3.11+, and frontend web applications in React 19 / TypeScript with Vite.

2. API Gateway & Communication

External and frontend-to-backend communication uses GraphQL and REST over HTTPS with strict OpenAPI / JSON schema validation. Inter-service backend communication uses gRPC with Protocol Buffers and Kafka message streams for asynchronous event-driven workflows.

3. Database & Caching Standards

Transactional persistence uses PostgreSQL 16 with read replicas and connection pooling via PgBouncer. Caching layers utilize Redis clusters with defined TTL policies. Vector search is powered by ChromaDB / pgvector.

4. Infrastructure & Container Orchestration

All workloads run as Docker containers orchestrated via Kubernetes (EKS/GKE). Infrastructure is provisioned strictly via Terraform (Infrastructure as Code).

6. Service Ownership

Every microservice has a single owning team recorded in CODEOWNERS and the service catalog. Owners are accountable for runbooks, on-call coverage, dependency upgrades, and deprecation notices.

7. API Versioning and Deprecation

Breaking API changes require a new versioned endpoint. Deprecated endpoints carry Sunset headers and a minimum 90-day notice before removal, announced in the engineering changelog.

8. Observability Standards

Services emit structured logs (PII-masked per the Data Privacy policy), RED metrics, and distributed traces into Datadog / CloudWatch. Every service ships with dashboards and alert thresholds reviewed quarterly.